Testing Hubble’s Law with a Small Sample of Nearby Galaxies

An Exploratory Velocity–Distance Analysis Using Published Cepheid-Calibrated Distances

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2026

Abstract

This project tests a simple velocity–distance relation using 15 nearby galaxies with published redshifts and Cepheid-calibrated distance-modulus measurements. The mean individual v/D value is 66.41 km/s/Mpc, the median is 57.97 km/s/Mpc, and the sample standard deviation is 23.08 km/s/Mpc. An exploratory free-intercept ordinary least-squares regression gives v = 15.758D + 559.655, with $R^2 = 0.08378$. These numbers are not interpreted as a precision measurement of the universal Hubble constant. Instead, the project uses the small sample to examine how peculiar velocities, distance uncertainty, reference-frame choices, sample size, and regression assumptions affect a simple Hubble-law analysis.

1. Introduction

Hubble’s law is commonly introduced as v = H■D: at sufficiently large scales, recession velocity is approximately proportional to distance. The calculation looks simple, but measuring H■ observationally is not. The HST Key Project emphasized the difficulty of establishing accurate distances and treated local-flow corrections and systematic uncertainties as important parts of the problem. This project deliberately works at a much smaller scale: instead of attempting a precision cosmological measurement, it asks what happens when the textbook relation is applied directly to a small set of real nearby galaxies.

2. Research Question

How well does a simple velocity–distance relation describe a small sample of nearby galaxies, and what does the scatter tell us about estimating H■?

3. Data and Provenance

The analysis uses 15 galaxies selected from the 27 Cepheid-calibrated nearby galaxies listed in Oxford Academic Table 2 associated with Freedman et al. (2001). That table gives the galaxy identifier, redshift z, distance modulus DM, and its 1-sigma uncertainty; it identifies the redshift source as the NASA/IPAC Extragalactic Database (NED). The sample was kept intentionally small so that every row could be inspected directly. Using a common distance-calibration family also avoids mixing several unrelated distance indicators in the same beginner-level analysis.

Galaxy	z	DM	σDM
NGC 4258	0.001494	29.44	0.07
NGC 4548	0.001621	30.88	0.05
NGC 2541	0.001828	30.25	0.05
NGC 925	0.001845	29.80	0.04
NGC 3198	0.002202	30.68	0.08
NGC 4414	0.002388	31.10	0.05
NGC 3627	0.002425	29.86	0.08
NGC 3621	0.002435	29.08	0.06
NGC 3319	0.002465	30.64	0.09
NGC 3351	0.002595	29.85	0.09
NGC 7331	0.002722	30.81	0.09

Galaxy	z	DM	σDM
NGC 3368	0.002992	29.97	0.06
NGC 2090	0.003072	30.29	0.04
NGC 4639	0.003395	31.61	0.08
NGC 4725	0.004023	30.38	0.06

4. Calculations

For the low-redshift sample, velocity was calculated as v = cz, using c = 299,792.458 km/s. Distance was calculated from the distance modulus as D = 10^((DM – 25)/5) Mpc. The propagated distance uncertainty is σD = D[ln(10)/5]σDM. Finally, v/D was calculated for each galaxy as a descriptive H■-like diagnostic. A free-intercept, unweighted ordinary least-squares regression of v on D was used only as an exploratory description of the sample.

5. Results

Statistic	Value	Unit
Mean individual v/D	66.41	km/s/Mpc
Median individual v/D	57.97	km/s/Mpc
Sample SD	23.08	km/s/Mpc
Range	32.40–111.51	km/s/Mpc
OLS slope	15.758	km/s/Mpc
OLS intercept	559.655	km/s
R^2	0.08378	dimensionless

The galaxy-by-galaxy v/D values span a wide range. The free-intercept regression has a low R^2, meaning that this particular nearby sample does not trace a strong linear velocity–distance relation. Importantly, the slope is not H■: the model includes an intercept, and nearby observed velocities can contain substantial non-Hubble contributions.

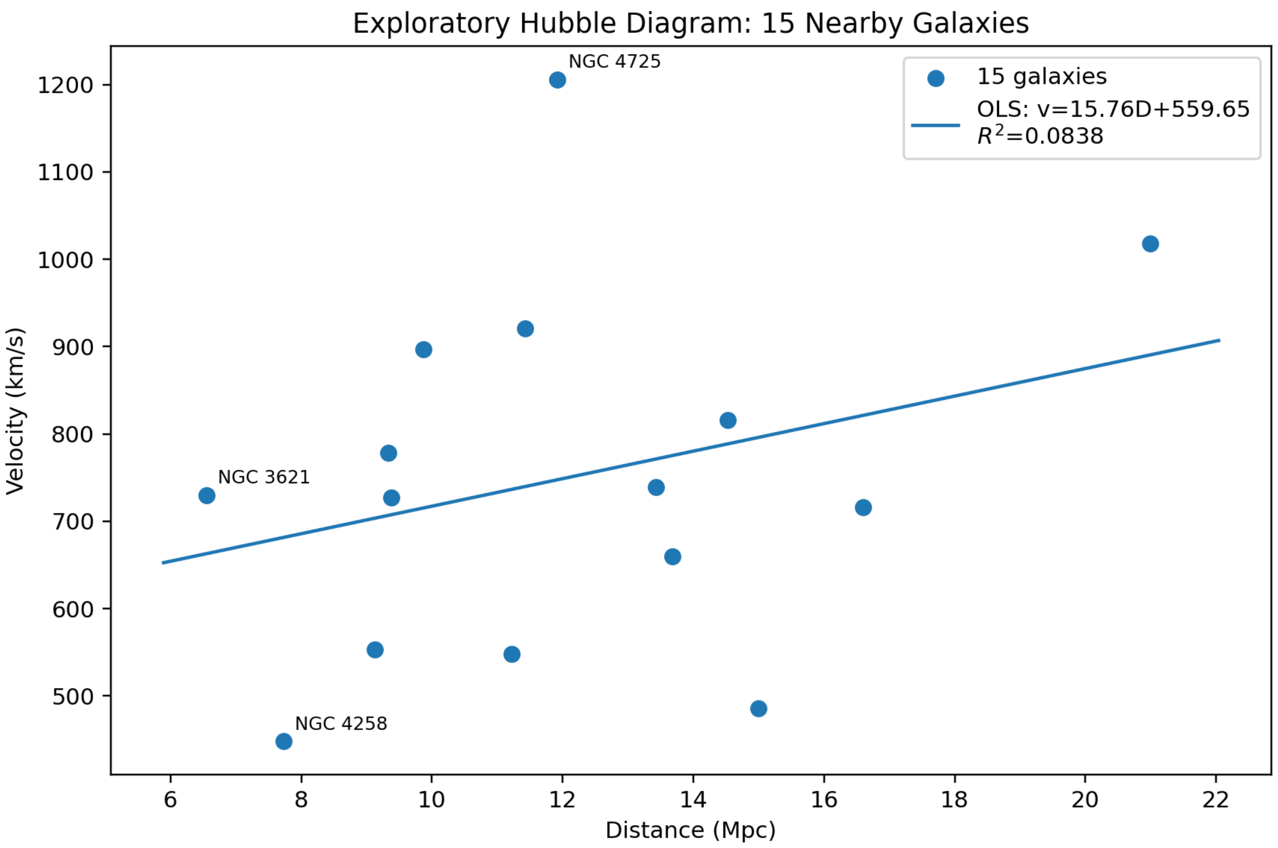

Figure 1. Exploratory Hubble diagram for the 15-galaxy sample.

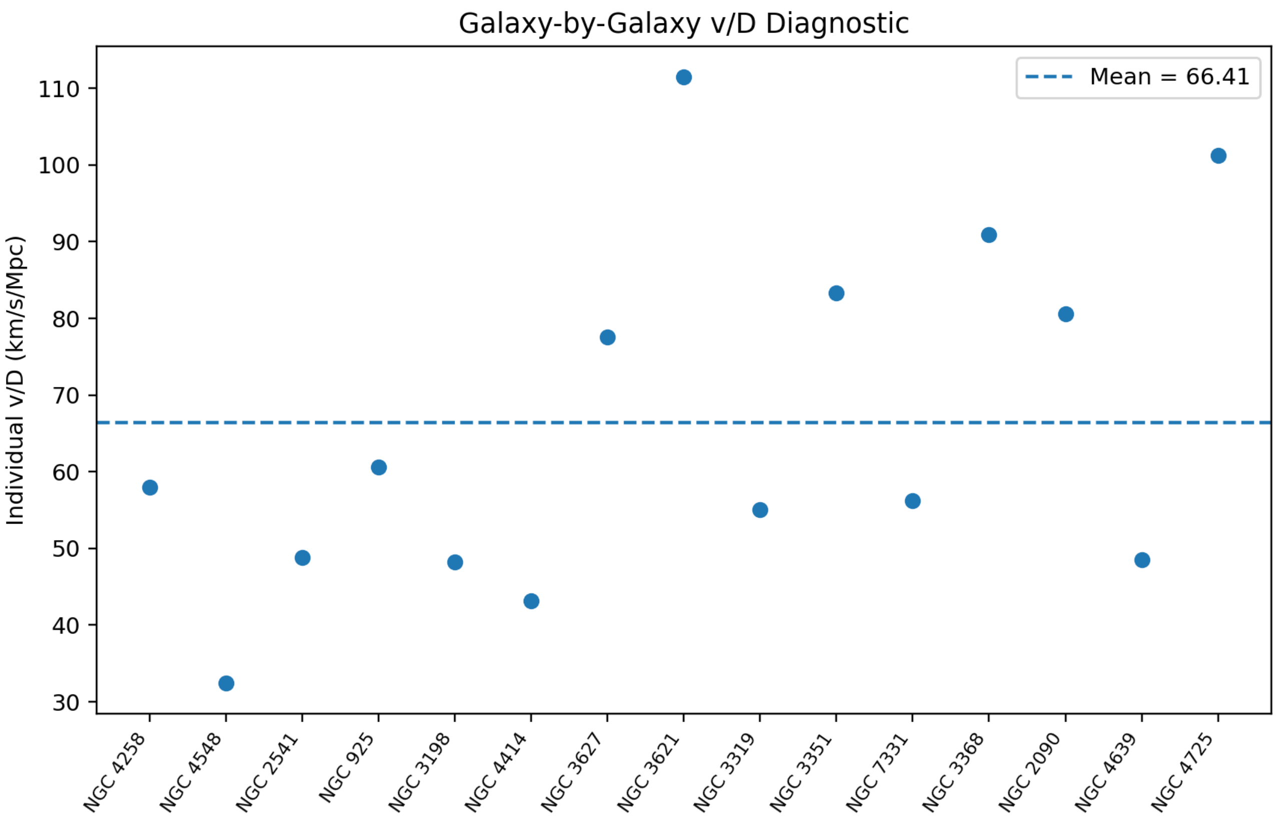

Figure 2. Individual v/D values. The dashed line is the descriptive sample mean.

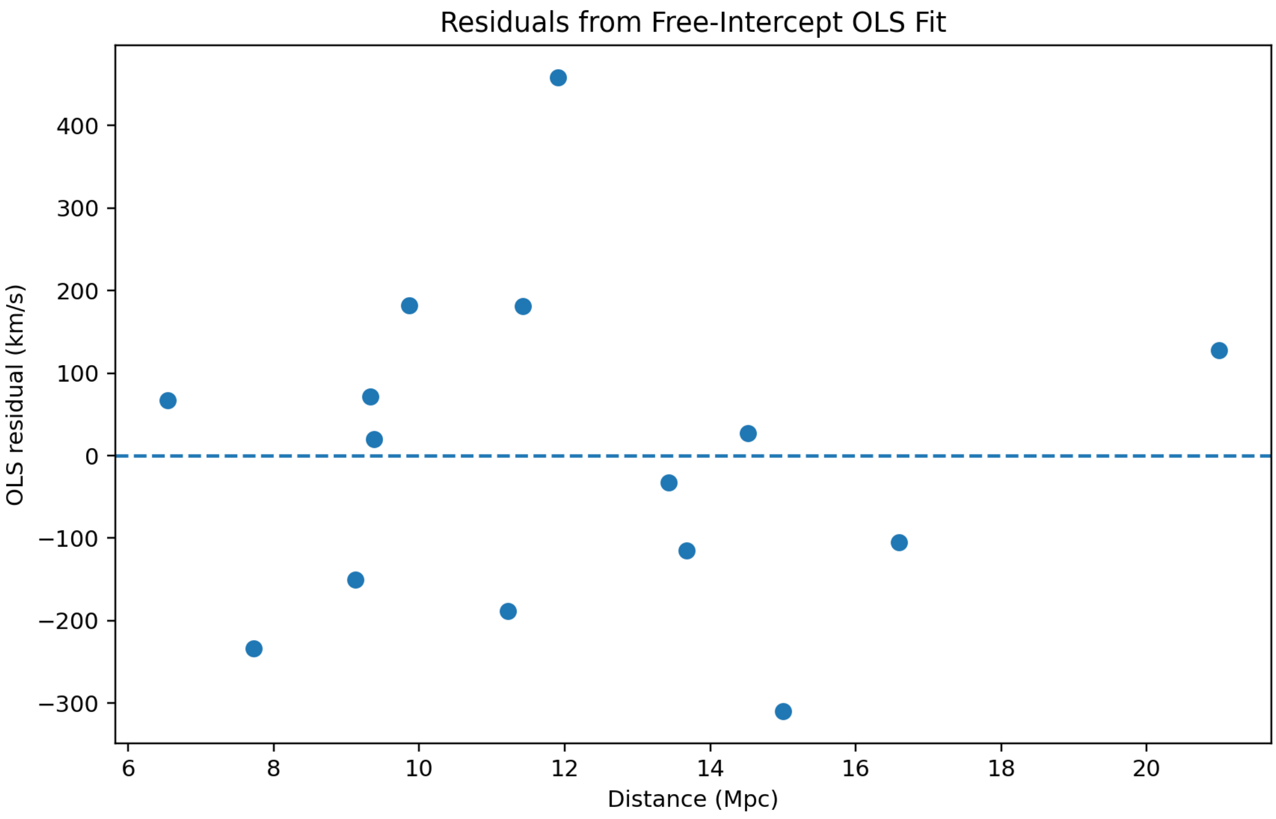

Figure 3. Residuals from the exploratory free-intercept OLS model.

6. Discussion

The weak relation is scientifically informative. These galaxies are nearby enough that peculiar velocities—motions caused by local gravitational structure rather than the smooth cosmic expansion—can be a significant fraction of the observed velocity. This can move an individual point above or below the ideal Hubble relation. The analysis also uses the redshifts as listed in the source table and does not apply a detailed local-flow or CMB-frame correction. Therefore the calculation should not be compared directly with a carefully flow-corrected Hubble diagram.

Distance uncertainty is another important effect. Because v/D depends inversely on distance, uncertainty in the distance modulus propagates directly into the individual ratios. The small sample also means that a few galaxies can strongly influence both the mean and the regression. Finally, the 15 distance measurements share a common calibration framework, so they should not be treated as perfectly independent measurements of an absolute scale.

7. Limitations

This project is intentionally exploratory. It uses only 15 galaxies, nearby objects, raw listed redshifts, and an unweighted free-intercept regression. It does not attempt a full flow model, a statistically complete survey analysis, or a hierarchical treatment of correlated systematics. These are not hidden omissions; they define the scope of the project. The analysis is meant to show how a simple equation behaves when applied to real measurements.

8. What I Learned

The most useful result was not a single value of H■. The analysis made clear that the hard part of observational cosmology is often deciding what a numerical result actually represents. A value can have several decimal places while still being limited by reference-frame choices, local motions, distance calibration, sample selection, and model assumptions. That distinction between numerical precision and physical reliability is the central lesson of this project.

9. Conclusion

For this 15-galaxy sample, the mean individual v/D is 66.41 km/s/Mpc, while the median is 57.97 km/s/Mpc and the sample standard deviation is 23.08 km/s/Mpc. The free-intercept OLS relation is $v = 15.758D + 559.655$, with $R^2 = 0.08378$. The weak fit should not be treated as a failure. It shows that a small nearby-galaxy sample is not a clean proxy for the global Hubble flow. A larger and more distant sample, with carefully chosen velocity-frame and flow corrections, would be the appropriate next scientific extension.

10. Reproducibility

The accompanying workbook retains the input data and formulas. The processed CSV and summary CSV are generated from the same 15 rows. The three figures are generated from the same calculations used in the report. The source notes document the published table and the NED provenance stated by that table. Exact object-by-object NED query URLs were not retained in the original materials, so none are invented here.

References

Freedman, W. L., et al. (2001). *Final Results from the Hubble Space Telescope Key Project to Measure the Hubble Constant.* The Astrophysical Journal, 553, 47–72. DOI: 10.1086/320638.

Oxford Academic. Table 2: *The 27 Cepheid-calibrated nearby galaxies published in Freedman et al. (2001).* https://academic.oup.com/view-large/129742892

NASA/IPAC Extragalactic Database (NED), California Institute of Technology. https://ned.ipac.caltech.edu/

Complete Project Appendix

Testing Hubble's Law with a Small Sample of Nearby Galaxies
This appendix is attached to the main research report so that the entire project can be submitted as one self-contained PDF.

1. Final audited results

Statistic	Final value
Mean individual v/D	66.41 km/s/Mpc
Median	57.97 km/s/Mpc
Sample standard deviation	23.08 km/s/Mpc
Range	32.40–111.51 km/s/Mpc
Free-intercept OLS	v = 15.758D + 559.655
R^2	0.08378

The mean v/D is a descriptive statistic for this sample, not a precision measurement of the universal Hubble constant. The free-intercept regression slope is likewise not identified with H■.

2. Complete 15-galaxy dataset

Galaxy	z	DM	σDM	v (km/s)	D (Mpc)	σD (Mpc)	v/D	Residual (km/s)
NGC 4258	0.001494	29.44	0.07	447.89	7.727	0.249	57.97	-233.52
NGC 4548	0.001621	30.88	0.05	485.96	14.997	0.345	32.40	-310.01
NGC 2541	0.001828	30.25	0.05	548.02	11.220	0.258	48.84	-188.44
NGC 925	0.001845	29.80	0.04	553.12	9.120	0.168	60.65	-150.25
NGC 3198	0.002202	30.68	0.08	660.14	13.677	0.504	48.27	-115.04
NGC 4414	0.002388	31.10	0.05	715.90	16.596	0.382	43.14	-105.27
NGC 3627	0.002425	29.86	0.08	727.00	9.376	0.345	77.54	19.60
NGC 3621	0.002435	29.08	0.06	729.99	6.546	0.181	111.51	67.18
NGC 3319	0.002465	30.64	0.09	738.99	13.428	0.557	55.03	-32.26
NGC 3351	0.002595	29.85	0.09	777.96	9.333	0.387	83.36	71.24
NGC 7331	0.002722	30.81	0.09	816.04	14.521	0.602	56.20	27.56
NGC 3368	0.002992	29.97	0.06	896.98	9.863	0.273	90.95	181.91
NGC 2090	0.003072	30.29	0.04	920.96	11.429	0.211	80.58	181.21
NGC 4639	0.003395	31.61	0.08	1017.80	20.989	0.773	48.49	127.39
NGC 4725	0.004023	30.38	0.06	1206.07	11.912	0.329	101.24	458.69

3. Reproducibility and calculation protocol

Velocity: v = cz, using c = 299,792.458 km/s.

Distance: D = 10^((DM – 25)/5) Mpc.

Distance uncertainty: σD = D ln(10)/5 × σDM.

Individual diagnostic: v/D.

Regression: ordinary least squares of velocity against distance with a fitted intercept.

The calculations are reproduced from the 15 raw input rows. The original project workbook is formula-driven, while this PDF preserves the final numerical outputs and dataset in a human-readable form.

4. Data provenance

Primary data table: Oxford Academic, Table 2, “The 27 Cepheid-calibrated nearby galaxies published in Freedman et al. (2001).” The table provides galaxy identifiers, redshift z, distance modulus, and 1-sigma distance-modulus uncertainty, and identifies NED as the source of the redshifts.

Original distance-scale context: Freedman, W. L., et al. (2001), “Final Results from the Hubble Space Telescope Key Project to Measure the Hubble Constant,” The Astrophysical Journal, 553, 47–72. DOI: 10.1086/320638.

NED: NASA/IPAC Extragalactic Database, California Institute of Technology. This project does not claim that the listed redshifts have been corrected into a CMB or flow-model frame.

Important provenance limitation: exact object-by-object NED query URLs were not retained in the supplied project materials. The project therefore documents the published table and its NED attribution rather than inventing individual query links.

5. Scientific interpretation

The low R^2 is treated as a meaningful result for this deliberately small nearby sample. Peculiar velocities, local gravitational structure, reference-frame choices, distance uncertainties, sample size, and the use of an unweighted regression all affect the appearance of a simple Hubble diagram. The analysis therefore demonstrates why the textbook relation is easier to write down than to measure observationally.

The project does not claim to solve the Hubble tension, determine the global H■, or provide a precision cosmological estimate. A larger and more distant sample with carefully corrected velocities and an appropriate statistical model would be a natural future extension.

6. Final project manifest

Component	Purpose
Main research report	Scientific narrative, methods, results, interpretation, limitations, conclusion and figures
Excel workbook	Formula-driven calculation and audit trail
Processed CSV	Machine-readable final dataset
Three figures	Hubble diagram, individual v/D diagnostic, residual plot
Summary statistics	Final numerical results
Source notes	Data provenance and reproducibility notes
README	Project overview and scope

Submission recommendation

Use this single complete PDF when a professor, mentor, counselor, or application reviewer wants one file that can be opened and read immediately. Keep the original ZIP/workbook privately as the reproducibility archive and use the full project folder for GitHub.